

# CS 486/686

# Unsupervised Learning

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Lecture 17

RN 21.7.1 · PM 10.2 · GBC 14.1, 20.10.4

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Learning

## Learning goals

- Understand what **unsupervised learning** is.
- Trace the **k-means** clustering algorithm.
- Perform **PCA**: mean-center, covariance, eigendecomposition.
- Understand the basic idea of **autoencoders** and **GANs**.

# Two big unsupervised tasks

No labels — only the raw data  $\{x_i\}$ . What can we learn?

## Representation learning



Find a low-dimensional summary of each example.

Clustering, PCA, autoencoders.

## Generative modelling



Learn the data distribution so we can *sample new examples*.

GANs, VAEs, diffusion.

Today: one task from each, plus two neural variants.

# Clustering: hard vs soft

Group training examples into **clusters**, treating them like discovered categories.

## Hard clustering

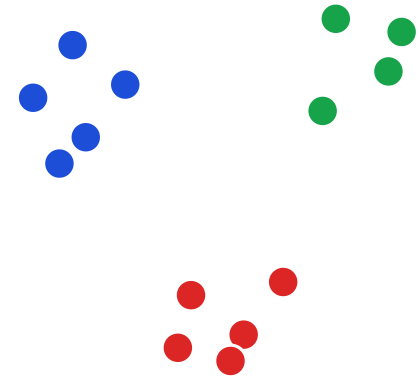
Each  $x$  gets  
*exactly one* cluster.

$$\text{class}(x) = c$$

## Soft clustering

Each  $x$  gets a  
*distribution* over  
clusters.

$$\text{class}(x) \sim P(C \mid x)$$



Each color is a discovered cluster — no labels were given.

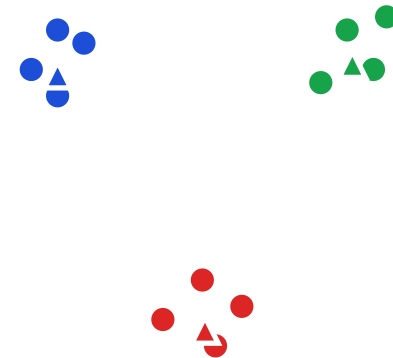
# k-Means: centroids and distance

**Inputs:** number of clusters  $k$  and  $m$  training examples  $x_i \in \mathbb{R}^n$ . Each cluster has a **centroid**  $c \in \mathbb{R}^n$ ; each  $x_i$  is assigned to its closest centroid.

## Euclidean distance (L2)

$$d(c, x) = \sqrt{\sum_{j=1}^n (c_j - x_j)^2}$$

- **Hard clustering:** each example goes to one centroid.
- Centroids live in the same feature space as the data.



Triangles = current centroids; circles = data points.

# The k-means algorithm

Alternate two steps until convergence:

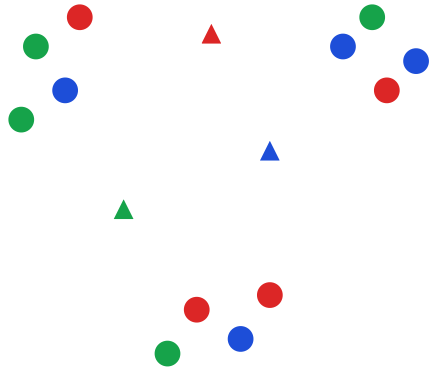
k-means( $X$ ,  $k$ )

1. Initialize  $k$  centroids  $C[0 \dots k - 1]$  randomly.
  2. Repeat until assignments stop changing:
    - **Assign step:** for each example  $i$ ,  $y[i] \leftarrow \arg \min_c d(C[c], X[i])$ .
    - **Update step:** for each cluster  $c$ ,  $C[c] \leftarrow$  mean of all  $X[i]$  with  $y[i] = c$ .
- Guaranteed to converge with L2 distance.
  - Not guaranteed to find the global optimum.
  - **Mitigate:** random restarts, feature scaling.

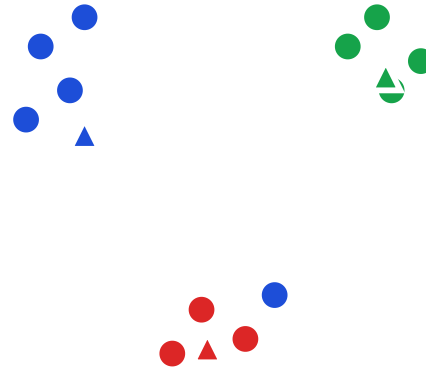
# k-Means in action

Same 12 points, three snapshots: random init → mid-iteration → converged.

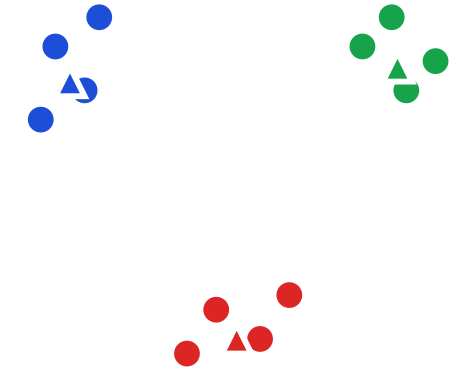
Iteration 0 (random init)



Iteration 2 (mid)



Converged



Centroids drift; assignments stabilize; cost decreases monotonically.

# One iteration: a numeric walk-through

$k = 2$ , Euclidean (squared) distance. Data + initial centroids:

| example | $x_1$ | $x_2$ | $x_3$ |
|---------|-------|-------|-------|
| 1       | 0.2   | 0.5   | 0     |
| 2       | -0.6  | 2.1   | 1.2   |
| 3       | -0.5  | 1.9   | 1.3   |
| 4       | 0.1   | 0.5   | -0.3  |

$$c_1 = (0.3, 0.8, -0.5), \quad c_2 = (-0.1, -0.5, 1.0).$$

Assign step (squared distance to each centroid):

$$\text{ex 1: } d^2(c_1) = 0.35, \quad d^2(c_2) = 2.09 \rightarrow$$

$c_1$

$$\text{ex 2: } d^2(c_1) = 5.39, \quad d^2(c_2) = 7.05 \rightarrow$$

$c_1$

$$\text{ex 3: } d^2(c_1) = 5.09, \quad d^2(c_2) = 6.01 \rightarrow$$

$c_1$

$$\text{ex 4: } d^2(c_1) = 0.17, \quad d^2(c_2) = 2.73 \rightarrow$$

$c_1$

Update step:

$$c'_1 = (-0.2, 1.25, 0.55), \quad c'_2 = \emptyset$$

**Watch out:**  $c_2$  got no examples. Re-initialize it (random restart).

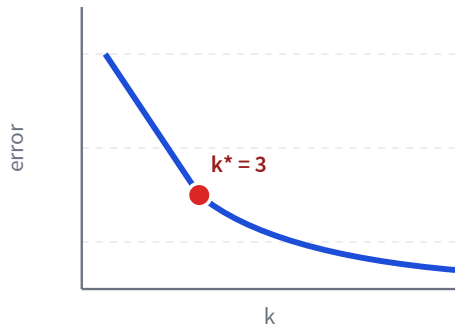


# How do we choose $k$ ?

Larger  $k$  always gives lower training error — but defeats the point of compression.

## Elbow method

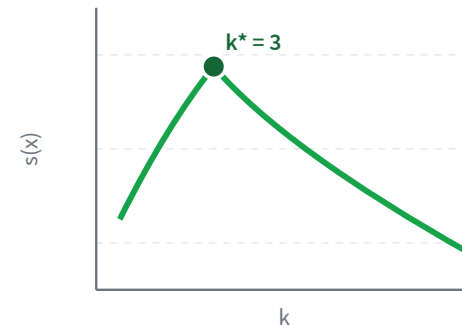
1. Run k-means for  $k \in \{1, \dots, k_{\max}\}$ .
2. Plot average within-cluster distance.
3. Pick the "elbow" where error stops dropping fast.



## Silhouette score

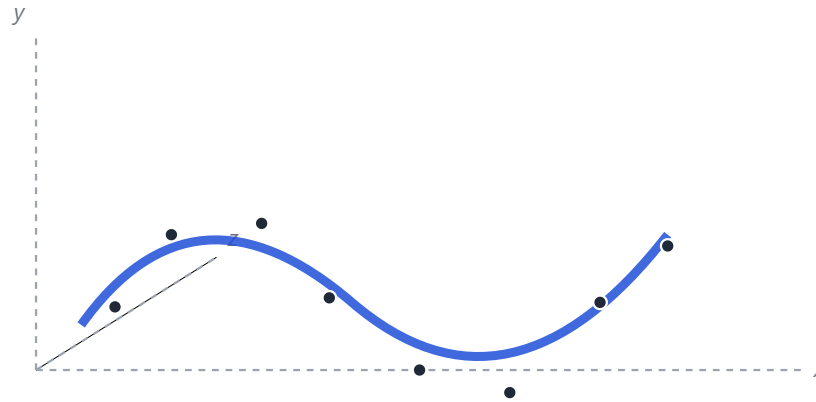
$$s(x) = \frac{b(x) - a(x)}{\max(a(x), b(x))}$$

- $a(x)$  = mean distance to own cluster.
- $b(x)$  = mean distance to *nearest other* cluster.
- Pick  $k$  maximizing average  $s$ .



# Dimension reduction: the manifold idea

High-dimensional data often lives on a **low-dimensional manifold**. Extra dimensions are mostly noise.



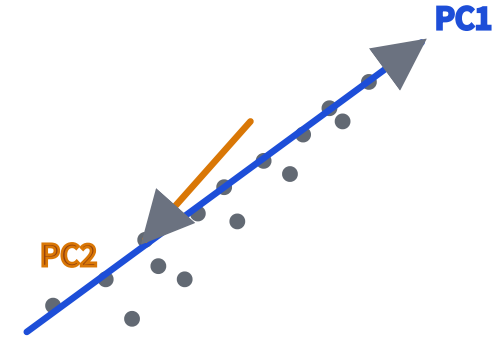
Goal: recover the intrinsic 1D coordinate along the blue ribbon.

# Principal Component Analysis (PCA)

## The PCA recipe

Find **orthogonal** axes that successively **maximize the variance** of the projected data.

- First PC = direction of largest variance.
- Second PC = direction orthogonal to first that maximizes remaining variance.
- Project onto the top  $k$  PCs for a compact representation.



# How to compute PCA

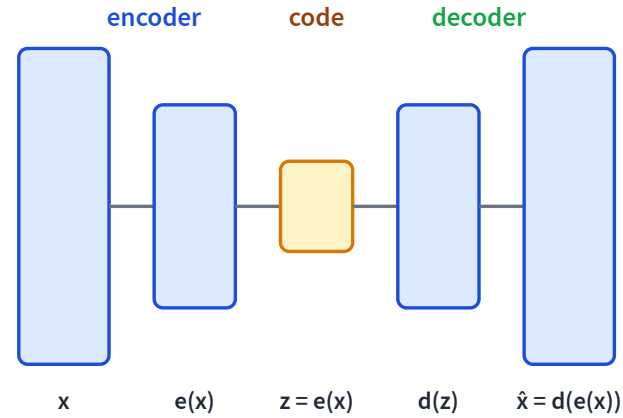
$$\text{PCA}(X, k)$$

1. Mean-center the data:  $x \leftarrow x - \text{mean}(x)$ .
2. Compute the covariance matrix  $\Sigma = \frac{1}{m} X^\top X$ .
3. Eigendecompose  $\Sigma$ :  $\Sigma v_k = \lambda_k v_k$ . Eigenvectors  $v_k$  are the principal components, sorted by  $\lambda_1 \geq \lambda_2 \geq \dots$
4. Project onto the top  $k$  PCs:  $z = V_{\text{top-}k}^\top x$ .

Fraction of variance kept by the  $k$ -th PC:  $\frac{\lambda_k}{\sum_j \lambda_j}$ . Cumulative across top  $k$  PCs  
guides how many to keep.

# Autoencoders: encoder + decoder

A neural net learns to **squeeze and reconstruct** its input.



**Loss:**

$$E = \sum_i (x_i - d(e(x_i)))^2$$

Encoder + decoder are trained jointly so that the bottleneck  $z$  keeps enough info to rebuild  $x$ .

# Linear vs deep autoencoders

## Linear autoencoder

Single linear layer; weights shared between encoder and decoder.

$$\hat{z} = Wx, \quad \hat{x} = W^\top \hat{z}$$

With squared loss, the solution recovers the same subspace as PCA.

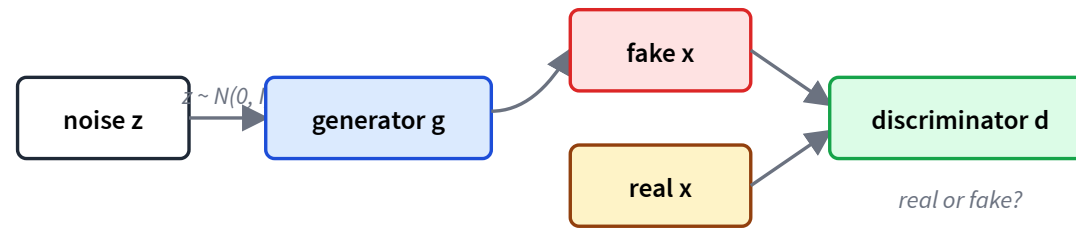
## Deep neural autoencoder

Stack non-linear layers (ReLU, etc.) in both encoder and decoder; train end-to-end with backprop.

Captures non-linear manifolds — the modern workhorse for representation learning.

# Generative Adversarial Networks (GANs)

Two networks play a game: a **generator** tries to fool a **discriminator**.



- **Generator  $g(z)$** : turns Gaussian noise  $z$  into a synthetic example that should look real.
- **Discriminator  $d(x)$** : binary classifier — was  $x$  drawn from training data, or from  $g$ ?

# GAN training: a minimax game

Both networks optimize the same value, in opposite directions:

$$E = \mathbb{E}_{x \sim p_{\text{data}}}[\log d(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - d(g(z)))]$$

## Discriminator

Maximize  $E$ : push  $d(\text{real}) \rightarrow 1$  and  $d(\text{fake}) \rightarrow 0$ .

## Generator

Minimize  $E$ : generate fakes that *fool* the discriminator.

At equilibrium:  $g$  produces realistic samples and  $d(x) \approx \frac{1}{2}$  — maximally uncertain.



## Learning goals (recap) — Next: decision trees

- ✓ Understand what **unsupervised learning** is.
- ✓ Trace the **k-means** clustering algorithm.
- ✓ Perform **PCA** by mean-centering, covariance, and eigendecomposition.
- ✓ Understand the basic idea of **autoencoders** and **GANs**.

L18: back to supervised — the **decision tree** learner.